

Module 5 Activity 3: Working with Neighbourhood Functions

Purpose:

Incorporate the **arcpy.sa** module to work with neighborhood functions to generate new raster datasets and analyze the results.

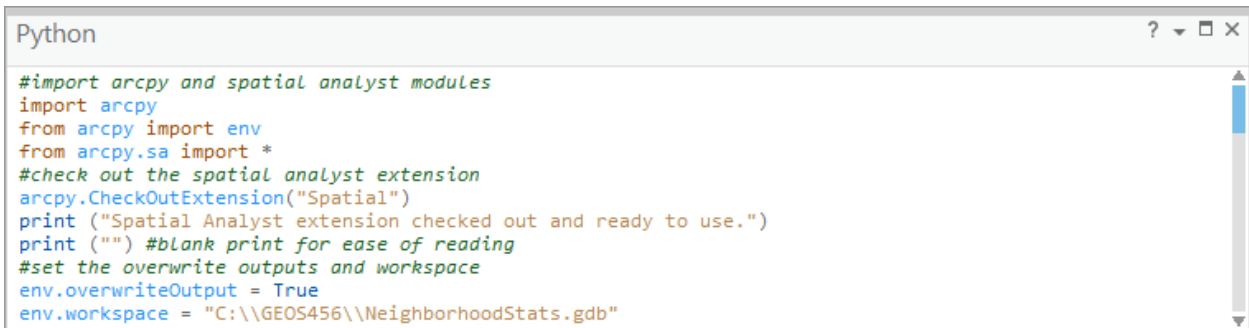
Instructions:

Any Python environment will work for this exercise.

Download the data **Module 5 Activity 3** and save the data to your working directory.

1. Import the **arcpy** module
2. Import the **arcpy.sa** module
3. Check out the spatial analyst extension so it can be used
4. Add a **print** statement to the **CheckOutExtension** function to notify the user that it is ready for use
5. Set the workspace and overwrite outputs

There will be additional blank **print** statements used throughout the script. This is to make the results easy to read. They won't be part of the process steps outlined in this activity and are completely optional.



```
Python
#import arcpy and spatial analyst modules
import arcpy
from arcpy import env
from arcpy.sa import *
#check out the spatial analyst extension
arcpy.CheckOutExtension("Spatial")
print ("Spatial Analyst extension checked out and ready to use.")
print ("") #blank print for ease of reading
#set the overwrite outputs and workspace
env.overwriteOutput = True
env.workspace = "C:\\GEOS456\\NeighborhoodStats.gdb"
```

6. Create a list of raster datasets present in the geodatabase
7. Describe the raster properties of each raster
 - Format
 - Cell size
 - Spatial reference

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print ("") #blank print for ease of reading
#set the overwrite outputs and workspace
env.overwriteOutput = True
env.workspace = "C:\\GEOS456\\NeighborhoodStats.gdb"
#get a list of the rasters in the geodatabase
rasterlist = arcpy.ListRasters()
#start a for loop to iterate through the list
for rasters in rasterlist:
    print ("Raster:", rasters)
#describe the raster properties
desc = arcpy.Describe(rasters)
#describe the spatial reference of the raster
spatialref = desc.spatialReference
#print all the raster properties
print ("") #blank print for ease of reading
print ("The format is:", desc.format)
print ("The cell size width is:", desc.meanCellWidth)
print ("The cell size height is:", desc.meanCellHeight)
print ("The coordinate system is:", spatialref.name)
print ("") #blank print for ease of reading
```

The next steps will use the **Neighborhood** toolset to generate a new raster with the **Block Statistics** tool using a **3x3 rectangle**

8. Create a variable called **landraster** that will hold the raster dataset to be used
9. Create another variable called **nbr** (neighborhood) to specify the parameters of the tool
10. Use the **Block Statistics** tool generate a new raster
11. Save the raster to the geodatabase in your workspace called **VarIndblock3_3**
12. Use a **print** statement to inform the user that the new raster has been created

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#set the overwrite outputs and workspace
env.overwriteOutput = True
env.workspace = "C:\\GEOS456\\NeighborhoodStats.gdb"
#get a list of the rasters in the geodatabase
rasterlist = arcpy.ListRasters()
#start a for loop to iterate through the list
for rasters in rasterlist:
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print ("The format is:", desc.format)
print ("The cell size width is:", desc.meanCellWidth)
print ("The cell size height is:", desc.meanCellHeight)
print ("The coordinate system is:", spatialref.name)
print ("") #blank print for ease of reading
#use the lancover raster in the neighborhood function
landRaster = "lndcover"
nbr = NbrRectangle(3,3,"CELL")
#create a raster using BLOCK STATS from the neighborhood toolset
#use all default settings for optional parameters
OutBlockStat = BlockStatistics(landRaster, nbr, "VARIETY")
#save the raster to your workspace
OutBlockStat.save("Varlndblock3_3")
#print result
print ("Block stats with 3x3 VARIETY neighbourhood created.")
print ("") #blank print for ease of reading
```

The results have been generated. The new raster is an integer raster dataset, meaning there is an attribute table associated with it. Next, we will create a list of the fields and access the values from the list.

13. Create a list of fields
14. Use a **print** statement to return the field names
15. Use a **SearchCursor** to search through the attributes and retrieve the values of the fields specified in the cursor. In this case, only the **Value** and **Count** fields

NOTE: call the raster dataset directly in the cursor, rather than referencing the variable

16. Start a **for** loop and the **getValue** function to return the values of the fields
17. Check the spatial analyst extension back in to the license manager

```
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import arcpy
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from arcpy.sa import *
#check out the spatial analyst extension
arcpy.CheckOutExtension("Spatial")
print ("Spatial Analyst extension checked out and ready to use.")
print ("") #blank print for ease of reading
#set the overwrite outputs and workspace
env.overwriteOutput = True
env.workspace = "C:\\GEOS456\\NeighborhoodStats.gdb"
#get a list of the rasters in the geodatabase
rasterlist = arcpy.ListRasters()
#start a for loop to iterate through the list
for rasters in rasterlist:
    print ("Raster:", rasters)
#describe the raster properties
desc = arcpy.Describe(rasters)
#describe the spatial reference of the raster
spatialref = desc.spatialReference
#print all the raster properties
print ("") #blank print for ease of reading
print ("The format is:", desc.format)
print ("The cell size width is:", desc.meanCellWidth)
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landRaster = "lndcover"
nbr = NbrRectangle(3,3,"CELL")
#create a raster using BLOCK STATS from the neighborhood toolset
#use all default settings for optional parameters
OutBlockStat = BlockStatistics(landRaster, nbr, "VARIETY")
#save the raster to your workspace
OutBlockStat.save("Varlndblock3_3")
#print result
print ("Block stats with 3x3 VARIETY neighbourhoo created.")
print ("") #blank print for ease of reading
#list the fields of the raster using a for loop
fieldList = arcpy.ListFields(OutBlockStat)
#for field in fieldList:
print ("Field name: " + field.name)
#use a search cursor to retrieve values from the attribute fields
SCursor = arcpy.da.SearchCursor("Varlndblock3_3",["Value","Count"])
print ("") #blank print for ease of reading
#use a for loop to iterate through the fields
for row in SCursor:
    #print the results for easy reading
    print ("Value: ", row[0], " ", "Count:", row[1])
#check out the spatial analyst extension
arcpy.CheckInExtension("Spatial")
print ("") #blank print for ease of reading
print ("Spatial Analyst returned to license manager.")
```

18. Run the script and examine you results

```
Python
Spatial Analyst extension checked out and ready to use.

Raster: cm_dem
Raster: lndcover
Raster: Varlndblock3_3

The format is: FGDBR
The cell size width is: 25.0
The cell size height is: 25.0
The coordinate system is: NAD_1983_UTM_Zone_12N

Block stats with 3x3 VARIETY neighbourhood created.

Field name: Count

Value: 1    Count: 116172.0
Value: 2    Count: 33939.0
Value: 3    Count: 2349.0
Value: 4    Count: 54.0

Spatial Analyst returned to license manager.
```

Answer the following questions:

1. How do you interpret the results of the block statistics?
2. What does **VARIETY** refer to in the context of grid statistics?

Next Steps:

Modify your script to include different parameters, such as modifying the search radius to **5x5**, or modifying the statistics type to **MAJORITY**.

Repeat the activity steps with the **Focal Statistics** tool and compare your results.